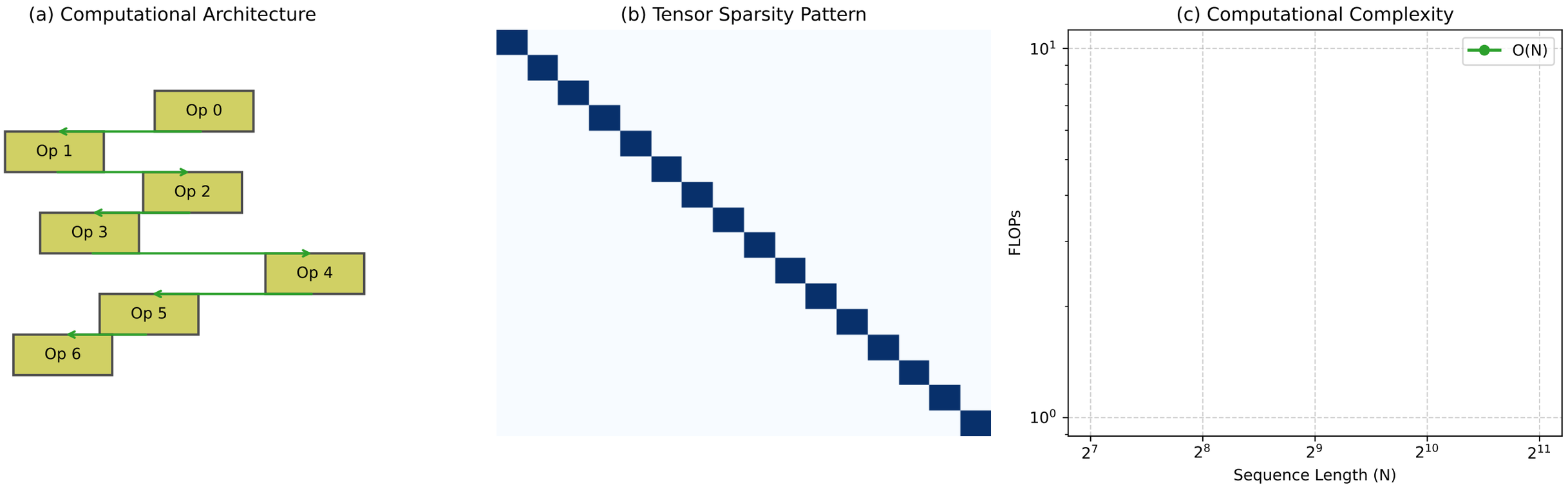Module Architecture: utils